

PCAccess — Full Technical Presentation

Enterprise Remote PC Access, Bi-Directional Streaming & Virtual Drive Ecosystem

English Edition

Architecture & Workflow

Tech Stack: .NET 4.8 + SignalR

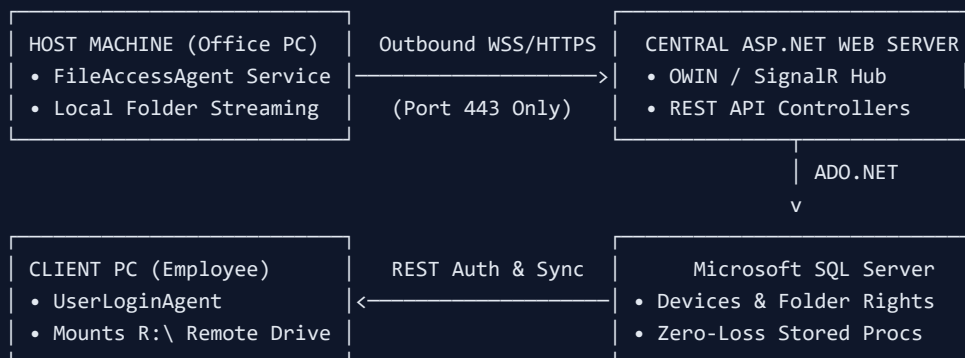
Zero-Trust Outbound

1. Executive Summary

PCAccess is an enterprise remote PC access and distributed file management platform that allows companies to connect on-premise workstations and shared directories to authenticated users globally without complex VPN setups, static public IPs, or router firewall modifications.

Key Architecture: The web portal (<https://pccaccess.ourdemos.com>) provides central administrative control, while desktop employees access shared files seamlessly through a locally mounted **R: Virtual Drive** in Windows Explorer.

2. High-Level System Architecture



3. Technology Stack Inventory

Layer / Component	Technology	Purpose & Technical Value
Web Backend	ASP.NET MVC 5 (.NET Framework 4.8)	Enterprise-grade controllers, session security, robust view rendering on Windows Server / IIS.
Real-Time Pipeline	Microsoft SignalR 2.4.3 (OWIN)	Full-duplex WebSocket hub managing device heartbeats, instant disconnect detection, and file streams.
Host Desktop Agent	C# Console / Windows Service (FileAccessAgent)	24x7 background folder provider with automatic 6-digit PIN handshake.
Client Desktop Agent	C# Console Client (UserLoginAgent)	Interactive desktop app that authenticates users and mounts virtual drive R: via Win32 kernel calls.
Database Layer	Microsoft SQL Server + Pure Stored Procs	Zero raw SQL in C#. Enforces role-based permissions, transactional integrity, and connection pooling.

4. Complete Business Workflow

- 1 Device Registration:** Office PC launches FileAccessAgent; Administrator inputs the 6-digit code on the Web Portal.
- 2 Folder Permissioning:** Administrator assigns specific local folders to employees with View, Download, or Upload rights.
- 3 1-Click Desktop Mount:** Employee runs UserLoginAgent; drive R: is instantly created under "This PC".
- 4 Secure Native Browsing:** Employee opens files directly inside standard desktop apps (Word, Excel, CAD, Explorer).